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App Name - "PathoNexa" ex. (pathonexa.com , pathonexa.in , pathnexa.co.in)

Pathology Lab Management System (Android + Web)

Project Overview

A complete Lab Management System (LMS) for small and medium pathology labs. The software will generate professional PDF reports, manage patients, doctors, payments, commissions, expenses, and provide complete business analytics.

Tech Stack

Mobile App

- React Native

Admin Panel

- React.js

Backend

- Node.js + Express

Database

- MongoDB

Storage

- Cloud Storage (PDF)

Authentication

- JWT Authentication

User Roles

1. Super Admin

- Manage all labs

- Subscription management
- Templates
- Reports
- Revenue
- Analytics

2. Lab Owner

- Full access
- Patients
- Doctors
- Reports
- Payments
- Expenses
- Staff
- Settings

3. Receptionist

- Patient Registration
- Payment Entry
- Receipt Print

4. Technician

- Fill Report Values
- Edit Report
- Generate PDF

Complete Workflow

Patient Arrives

↓

Search Mobile Number

↓

If Patient Exists

↓

Open Patient History

Else



Create New Patient



Select Doctor



Select Test / Package



Auto Price



Discount



Paid Amount



Pending Amount



Payment Mode



Sample Collection



Technician Enters Report Values



Generate PDF



Owner Verification (Optional)



Print



WhatsApp Share



Report Saved in History

Dashboard

Display

- Today's Revenue
- Today's Reports
- Pending Reports
- Pending Payments
- Doctor Commission
- Monthly Revenue
- Total Expense
- Total Profit
- Recent Reports

Patient Module

Fields

- Patient ID (Auto Generate)
- Barcode
- QR Code
- Name

- Age
- Date of Birth
- Gender
- Mobile Number
- Alternate Mobile
- Address
- City
- State
- Pincode
- Email
- Blood Group
- Remarks
- Registration Date
- Report History

Features

- Patient Search
- Patient History
- Edit Patient
- Delete Patient

Doctor Module

Fields

- Doctor Name
- Clinic Name
- Specialization
- Address
- Mobile Number
- WhatsApp Number
- Commission Percentage
- Bank Details
- UPI ID
- Status
- Notes

Doctor Ledger

- Total Reports
- Total Commission

- Paid Commission
- Pending Commission
- Monthly Statement
- PDF Export

Test Master

Admin can create unlimited tests.

Fields

- Test Name
- Category
- Department
- Price
- Template
- Report Header
- Status

Every Test Contains

- Parameter Name
- Short Name
- Unit
- Normal Range
- Critical High
- Critical Low
- Decimal Places
- Male Range
- Female Range
- Child Range
- Print Order
- Bold
- Highlight

Example

CBC

- WBC
- RBC
- HGB

- MCV
- MCH
- MCHC
- Platelet
- etc.

Package Module

Admin can create packages.

Example

Full Body Checkup

Includes

- CBC
- LFT
- KFT
- Blood Sugar
- Thyroid
- Urine Routine
- Vitamin D
- Vitamin B12

Report Creation

Select Patient

↓

Select Doctor

↓

Select Test / Package

↓

Auto Fetch Price



Discount



Paid Amount



Pending Amount



Sample Collection Date



Report Date



Technician Name



Enter Report Values



Save



Generate PDF



Print



WhatsApp

↓

Download

Auto High / Low Detection

Technician only enters Result Value.

Example

Result = 5.7

Range = 4.0 - 10.0

Status = Normal

Example

Result = 15

Range = 4.0 - 10.0

Status = High (H)

Example

Result = 2

Range = 4.0 - 10.0

Status = Low (L)

System automatically calculates High/Low.

Technician never enters H/L manually.

Professional PDF

Header

- Lab Logo
- Lab Name
- Address
- Contact Number

Patient Details

- Name
- Age
- Gender
- Patient ID
- Ref By Doctor
- Report Date

Report Table

- Parameter
- Result
- Unit
- Normal Range
- Alert

Footer

- QR Code
- Barcode
- Signature
- Generated Date
- Paid / Unpaid Stamp

Buttons

- View
- Print
- Download
- WhatsApp
- Share
- Edit
- Delete

Payment Module

Fields

- Invoice Amount
- Discount
- Paid Amount
- Pending Amount
- Payment Mode
- Transaction ID
- Payment Date

Payment Modes

- Cash
- UPI
- Card
- Bank Transfer

Ledger

Every transaction automatically saved.

Doctor Commission Module

Example

Report Price = ₹500

Commission = 20%

Doctor Wallet = ₹100

Lab Revenue = ₹400

Doctor Payment

- Pay Commission
- Payment History
- Pending Commission

Expense Module

Expense Categories

- Electricity
- Rent
- Staff Salary
- Chemical
- Needle
- Syringe
- Tube
- Printer Ink
- Internet
- Other

Monthly Expense Report

Analytics

- Daily Revenue
- Weekly Revenue
- Monthly Revenue
- Yearly Revenue
- Total Profit
- Total Expense
- Pending Payments
- Doctor Wise Revenue
- Test Wise Revenue
- Most Performed Test

Report History

Search By

- Patient Name
- Mobile
- Patient ID
- Doctor
- Date
- Barcode

- Report Number

Actions

- View
- Print
- Download
- Share
- WhatsApp
- Edit
- Delete
- Duplicate

WhatsApp Module

One Click Report Share

Message Example

Hello {Patient Name},

Your pathology report is ready.

Please find your report attached.

Thank You.

Notifications

- Report Ready
- Payment Pending
- Subscription Expiry
- Doctor Commission Due

Backup

- Automatic Cloud Backup
- Manual Backup

- Restore Backup

Staff Management

Roles

- Owner
- Receptionist
- Technician
- Manager

Permission Based Access

Subscription System

- 7 Day Free Trial
- Monthly Plan
- Yearly Plan
- Expiry Reminder

Settings

- Lab Logo
- Lab Name
- Address
- Contact Number
- Email
- GST Number
- Footer
- Signature
- Stamp
- Theme
- Language

Admin Panel

Manage

- Patients
- Doctors
- Tests
- Templates
- Packages
- Reports
- Payments
- Expenses
- Staff
- Settings
- Subscription
- Analytics
- Backup

Database Collections

- Users
- Patients
- Doctors
- Tests
- TestTemplates
- Packages
- Reports
- ReportValues
- Payments
- DoctorLedger
- Expenses
- Staff
- Settings
- Notifications
- Subscriptions

Maked by - premveer

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